Quantifying Model Misrepresentation in Geophysical Inversion for Critical Mineral Exploration

Zhen Yin, Alex Miltenberger, Mark Topinka, Lijing Wang, Tapan Mukerji, and Jef Caers

***Abstract*— Geophysical inversion often leans on simplified geological models due to a lack of detailed geological information in greenfield critical mineral exploration. While the actual orebodies are more complex in geometry, the simplified model representation can introduce large uncertainties, thereby adversely affecting decisions for field development. We introduce a stochastic quantification and treatment of the model misrepresentation errors when inverting geophysical data. To test whether misrepresentation errors exist, we provide a generalized hypothesis testing approach. We start with the assumption that our current geometry model accurately represents the subsurface (null hypothesis). We then use geophysical data to test this assumption. If the null hypothesis is rejected, the geometry is known to be false. We employ a Bayes factor to quantify where in space the model error is most significant. The latter will be used as information to reparametrize the assumed geometry, to reduce the misrepresentation errors. Our approach also allows accounting for data error and model error jointly, by using a pushforward formulation of the inverse problem. We provide a sequential Monte Carlo (SMC) sampling algorithm to solve the pushforward formulation, enabling practical computation of the Bayes factor. Both the synthetic and real field studies showed the Bayes factor can locate where the model is misrepresented. The real field application also shows that, through mitigating the misrepresentation errors, we reduced the uncertainty in key decision-making parameters concerning the field development, including the prospecting conductors' volume and depth. We further provide open-source code to facilitate practical applications.**

***Index Terms*— Bayes factor, gravity, hypothesis test, sequential Monte Carlo (SMC), time-domain electromagnetic.**

I. Introduction

DETERMINING the geometry of perspective ore deposits is critical to decision-makings for mineral exploration and development, such as estimating economic potentials [1], [2], [3], planning drillholes [4], [5], geophysical surveys [6], [7], as well as the mining extraction [8], [9]. However, in greenfield mineral explorations, only limited information is available on the geometry of a prospective ore deposit. The same is true for critical mineral explorations, including cobalt, cooper, and nickels required for electrical vehicle (EV) batteries and the global energy transition [10]. Geology conceptual interpretations of intrusion models only provide a qualitative appraisal of what an orebody shape would look like [11] and [12]. Geophysical data inversion thus plays a significantly important role for quantifying the ore deposits geometries [13]. As a result, a large degree of freedom exists in inverting orebody geometries from geophysical data, and hence a large uncertainty exists.

Traditionally the deposit anomaly geometries are inferred by deterministic geophysical inversions using electromagnetic data [14], [15], gravity [16], magnetic data [17], [18], or jointly [19], [20]. First, deterministic inversion derives the orebody geometry by minimizing the difference between the observed geophysical data and model responses. However, solutions to the geophysical inverse problem are nonunique. Adding regularization terms in deterministic inversion often leads to an inverted body geometry that is much smoother than the true geometry. Second, deterministic inversion provides a limited representation of uncertainty [21]. To solve the problem, stochastic geophysical inversion is used to generate multiple realizations of geometrical models. For example, particle swarm optimization was used to infer the buried structure geometrical parameters such as depth and shape from gravity [22] and magnetic data [23]. Athens and Caers [25] proposed a Monte Carlo-based stochastic inversion of gravity data for complex geological structures with faults. Also with gravity inversion, Wei et al. [26] used Markov chain Monte Carlo (McMC) to sample multiple salt body shapes.

Most geophysical inversions, either deterministic or stochastic, use simple geometric shapes to represent the buried mineral anomalies when there are little other supportive data. This may well be sufficient if the goal is targeting the geophysical anomaly with drilling. Simple geometry has advantages of requiring only a small amount of parameters, thus making it easy for model parameterization [1]. Essa and Munschy [22] and Clark and Tonkin [27] used spherical shapes to represent the buried magnetic and gravity anomaly source body to perform geophysical inversion for real field mineral explorations [22]. Three-dimensional rectangular plates were used by McMillan et al. [28] to represent a thin dipping conductor when inverting the electromagnetic data over the Caber deposit in Quebec. Sun et al. [29] used uniform geometrical structures to infer 3-D orebodies using borehole

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and magnetic data. Three-dimensional cubes were used by Fournier et al. [30] to represent magnetic sources for Ni-Cu-PGE deposit for performing geophysical inversion using SimPEG. To represent the anomaly shapes in more complex geometries, some geophysical inversions used homogenous polyhedrons with straight lines [20], [29], [31]. Some other studies also attempt to represent complex anomaly shapes by combining several simple geometries. For example, Foss and McKenzie [32] combined 3-D plates with ellipsoids to model buried magnetic source bodies for inversions in Black Hill Norite from Australia. Essa and Elhussein [23] and Mehanee et al. [33] used sphere, cylinder, and thin-sheet models to invert gravity and magnetic data to infer buried anomaly body parameters including depth and locations for fields from Cana, USA, Sengal, and Egypt. Wei et al. [26] combined various ellipses through a concave hull to infer buried salt body shapes in gravity inversion. At the end of all inversions, the models that match the actual geophysical signals are accepted for making decisions, such as drill targeting.

The actual morphology of mineral deposits and their host intrusions are, however, much more complex. This is especially true for deposits of Ni–Cu–Co–PGE–hosted magmatic sulfides [12], [34], [35]. Using simple parametric shapes to approximate the complex geological morphology will thus result in model misrepresentations in geophysical inversions. As geophysical inversion is an ill-posed inverse problem, such approximation or misrepresentation can have detrimental effect on the inversion solutions and severely distort inverted model parameters [36], [37]. Decision-making needs to consider the uncertainty caused by the model misrepresentation. Few studies addressed the quantification of uncertainty stemming from the model misrepresentation errors. Calvetti et al. [36] treated model errors mainly from forward modeling as a random variable to quantify when solving inverse problems. We found nearly no study is available to quantify the model misrepresentations, fully and comprehensively, in geophysical inversion for mineral exploration applications, where models commonly require inherent assumptions or approximations.

To create a comprehensive stochastic treatment of the problem of model misrepresentation, we employ a hypothesis testing approach. We assume a particular geometry to represent the true subsurface geometry. This is the null hypothesis: there is no model misrepresentation error. Then we use a statistical test to attempt to reject that geometry as being valid. The test data is the geophysical data. If the null hypothesis is rejected, then the geometry is known to be false, with stated significance. When the null hypothesis is rejected, we employ a Bayes factor approach that allows quantifying where in 3-D space the model error is most significant. The latter will be used as information to reparametrize the assumed geometry. Our approach also allows accounting for separating the data error and model error, by using a push-forward formulation to the inverse problem. We provided a sequential Monte Carlo (SMC) inversion approach for computing the model errors, while the main proposed model error quantification can also be used in other alternative inversion methods given its general formulation. Our approach is validated on synthetic cases, where the model presentation is known. Then we applied it to a real case of inverting surface EM data to prospect for nickel-sulfide magmatic mineralization.

II. Methodology

A. Developing a Model Misrepresentation Test

Quantifying model error is challenging in situations where little or no prior knowledge about subsurface geometries is available. Model errors may occur because of several causes: the geology representation model (e.g., geometrical shape and parameterization) is wrong; the prior uncertainty on the parameterization is wrong (often too narrow); and the physical forward model is wrong, for example, the physics is too simple or lacking physical processes. In this work, we concentrate on the model misrepresentation errors, although our approach may also include the other components. Model misrepresentation is likely to be the largest error in cases where little prior information is available to perform any stochastic inversion.

The methodology we develop operates as follows: we start with some initial model representation (often a simple geometry), one which we apply the hypothesis test approach. Our approach to model error quantification will not only identify that the model is wrong (a binary statement) but also, if it is wrong, where the model is wrong. The latter will allow increasing the model complexity in a systematic way rather than an ad hoc attempt at many model parameterizations.

We design a hypothesis to test model misrepresentation errors. To show how it works, we will first review hypothesis testing, and how different formulations of the inverse problem are used to define such tests. Consider the following two hypotheses.

H_0: The current stated prior model is correct (representing the true geology).

H_1: An alternative prior model is correct.

Hypothesis testing requires data. Here the data is the geophysical data $\mathbf{d}_{\text{obs}}$. Our test will be formulated using a Bayes' factor (likelihood ratio), defined as the evidence ratio of the two hypotheses [38], [39]

$$B(H_0, H_1) = \frac{P(\mathbf{d}_{\text{obs}}|H_0)}{P(\mathbf{d}_{\text{obs}}|H_1)}. \tag{1}$$

$B(H_0, H_1)$ is the Bayes factor. $P(\mathbf{d}_{\text{obs}}|H)$ is the evidence probability under each hypothesis. If $B < 1$, the hypothesis H_0 is falsified (proven wrong). Using marginalization, the Bayes factor can be further developed as

$$B(H_0, H_1) = \frac{\int_{\Omega_{\mathbf{m}}} f(\mathbf{d}_{\text{obs}} \,|\, \mathbf{m}, H_0) f(\mathbf{m} \,|\, H_0) \mathrm{d}\mathbf{m}}{\int_{\Omega_{\mathbf{m}}} f(\mathbf{d}_{\text{obs}} \,|\, \mathbf{m}, H_1) f(\mathbf{m} \,|\, H_1) \mathrm{d}\mathbf{m}} \tag{2}$$

where $\mathbf{m}$ is the geological model parameters under hypothesis H, either H_0 or H_1. $\Omega_{\mathbf{m}}$ is the space of all the possible $\mathbf{m}$.

Calculation of the Bayes factor requires calculating the likelihood distribution $f(\mathbf{d}_{\text{obs}} \,|\, \mathbf{m}, H)$. In classical Bayesian formulation, however, the likelihood contains both the model and data error. Stating the likelihood is known to be a considerable challenge in Bayesian inverse models [40], [41], [42]. In our stated problem, the Bayesian likelihood formulation

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is not immediately practical because we aim to determine a model error that is significantly larger than the measurement error. Instead, to separate the model errors from the measurement error, we opt for a pushforward formulation of the inverse problem [43], [44]

$$f_{\text{post}}(\mathbf{m}) = \frac{\rho_{\text{obs}}(\mathbf{d})}{\rho_{\text{push}}(\mathbf{d})} f_{\text{prop}}(\mathbf{m}) \tag{3}$$

where $\mathbf{d} = g(\mathbf{m})$ is the modeled geophysical data from applying geophysical forward model $g(.)$ on $\mathbf{m}$. The pushforward formulation uses a proposed distribution of $\mathbf{m}$: $f_{\text{prop}}(\mathbf{m})$, like the prior in Bayes. However, it contains two other distributions.

1) $\rho_{push}(.)$: It is called the pushforward distribution of geophysical data $\mathbf{d}$. This distribution is the mapping of a distribution of geological model $\mathbf{m}$ into the distribution of data variables $\mathbf{d}$. To calculate a sample of the distribution ρ_{push}, one would first sample $\mathbf{m}^{(\ell)} \sim f_{\text{prop}}$ then run the forward model $\mathbf{d}^{(\ell)} = g(\mathbf{m}^{(\ell)})$. $\mathbf{d}^{(\ell)}$ obtained is thus regarded as a sample of ρ_{push}.
2) $\rho_{obs}(.)$: The measurement error distribution states, given the observations, what all possible observation could be because of lack of accuracy in the measurement device. A common choice for $\rho_{\text{obs}}(.)$ is a Gaussian probability density function (pdf) centered at mean $\mathbf{d}_{\text{obs}}$ with error covariance Σ_{err} such that $\rho_{\text{obs}}(.) = \mathcal{N}(\mathbf{d}_{\text{obs}}, \Sigma_{\text{err}})$. The mean and covariance can be estimated using repeated measurements, which we will show in the real case study.

The key element that makes this pushforward formulation [see (3)] attractive is that the measurement error is a numerator, while any model error in the denominator, meaning, there is a separation between model and data error, and that is not as easy to achieve using the classical Bayesian formulation. Using the notation from the pushforward formulation, the Bayes factor of (2) can be reformulated as follows (see Appendix A):

$$B(H_0, H_1) = \frac{\int_{\Omega_{\mathbf{d}}} \rho_{\text{push}}(\mathbf{d} \mid H_0)\rho_{\text{obs}}(\mathbf{d})\mathrm{d}\mathbf{d}}{\int_{\Omega_{\mathbf{d}}} \rho_{\text{push}}(\mathbf{d} \mid H_1)\rho_{\text{obs}}(\mathbf{d})\mathrm{d}\mathbf{d}}. \tag{4}$$

The Bayes factor requires calculating integrals over probability distributions. We can then approximate the calculation according to the law of the unconscious statistician [45], using Monte Carlo samples of the posterior $\mathbf{d}$

$$\int_{\Omega_{\mathbf{d}}} \rho_{\text{push}}(\mathbf{d} \mid H)\rho_{\text{obs}}(\mathbf{d})\mathrm{d}\mathbf{d} \cong \frac{1}{N} \sum \rho_{\text{push}}\left(\mathbf{d}_{\text{obs}}^{(\ell)} | H\right) \tag{5}$$

where $\mathbf{d}_{\text{obs}}^{(l)} \sim \rho_{\text{obs}}(.)$. And $\ell = 1, 2, \ldots, L$, is the index of discrete Monte Carlo samples. With (5), the Bayes factor becomes much easier to calculate

$$B(H_0, H_1) = \frac{\sum \rho_{\text{push}}\left(\mathbf{d}_{\text{obs}}^{(\ell)} | H_0\right)}{\sum \rho_{\text{push}}\left(\mathbf{d}_{\text{obs}}^{(\ell)} \mid H_1\right)}. \tag{6}$$

Because H_1 is the alternative hypothesis that correctly represents the geological truth, $\rho_{\text{push}}(\mathbf{d}_{\text{obs}} \mid H_1)$ should equal to $\rho_{\text{obs}}(\mathbf{d}_{\text{obs}})$: $\rho_{\text{push}}(\mathbf{d}_{\text{obs}} \mid H_1) = \rho_{\text{obs}}(\mathbf{d}_{\text{obs}})$. The Bayes factor thus can be further simplified as

$$B(H_0, H_1) = \frac{\sum \rho_{\text{push}}\left(\mathbf{d}_{\text{obs}}^{(\ell)} | H_0\right)}{\sum \rho_{\text{obs}}\left(\mathbf{d}_{\text{obs}}^{(\ell)}\right)}. \tag{7}$$

It is important to stress that $\rho_{\text{push}}(.)$ should be the final pushforward distribution from $f_{\text{post}}(\mathbf{m})$ and optimally match $\rho_{\text{obs}}(.)$. Only in this way, the model error can be tested with the inversion model parameters under the current geological hypothesis. $\rho_{\text{push}}(.)$ can be calculated using the Monte Carlo samples of the posterior data $\mathbf{d}^{(\ell)}$.

1) Take a posterior model sample $\mathbf{m}^{(\ell)} \sim f_{\text{post}}(\mathbf{m})$.
2) Pushforward $\mathbf{d}^{(\ell)} = g(\mathbf{m}^{(\ell)})$.
3) Estimate the final $\rho_{\text{push}}(.)$ using Monte Carlo samples of $\mathbf{d}^{(\ell)}$.

In Section II-B, we will provide an SMC for sampling the posterior data $\mathbf{d}^{(\ell)}$ such that $\rho_{\text{push}}(.)$ can be calculated.

$B(H_0, H_1)$ is the ratio between the pushforward and measurement error distributions evaluated in the observed data. Any wrong posterior will therefore create an imbalance in the ratio of (7). Because, in our problem setting, ρ_{obs} is very well-known, while ρ_{push} much less known (due to unknown model error), any imbalance in the ratio is attributed to a misrepresentation error of the model. As the formulation shows, the Bayes factor is a single scalar upon which a binary decision is made (wrong/not wrong). However, most geophysical data is either spatial, temporal, or a combination of both. For example, the surface EM data considered in the following case study is both spatial (locations of measuring station $\boldsymbol{x}$) and temporal (decay of the signal overt time t), hence, the Bayes factor can be written as follows:

$$B(\boldsymbol{x}, t) = \frac{\sum \rho_{\text{push}}\left(\mathbf{d}_{\text{obs}}^{(\ell)}(\boldsymbol{x}, t) | H_0\right)}{\sum \rho_{\text{obs}}\left(\mathbf{d}_{\text{obs}}^{(\ell)}(\boldsymbol{x}t)\right)}. \tag{8}$$

This means that the Bayes' factor informs in space ($\boldsymbol{x}$) and time (t) where the ratio becomes imbalanced.

B. SMC-ABC

The formulation of Bayes factor calls for Monte Carlos sampling of $\mathbf{d}^{(\ell)}$ from the posterior distribution to calculate $\rho_{\text{push}}(.)$. We will need to "solve" the geophysical inverse problem to get the posterior distribution $f_{\text{post}}(\mathbf{m})$, and hence running the pushforward. Again, we should also note that assuming a simple model with H_0, the inversion problem will not be solved yet. However, learnings will be made about where and when the model error is largest, which is the goal.

This proposed formulation of model misrepresentation error can use posterior samples from any stochastic geophysical inversion. Here, we use an SMC sampler to perform the geophysical inversion for sampling posterior $\mathbf{d}^{(\ell)}$. The concept and implementation of the SMC are summarized in Fig. 1 and Algorithm 1. In practice, SMC starts with Monte Carlo simulation on a user-defined simple prior distribution $f_1(\mathbf{m})$ (e.g., a wide uniform distribution), generating L model realizations $\mathbf{m} = \{\mathbf{m}^{(1)}, \ldots, \mathbf{m}^{(l)}, \ldots, \mathbf{m}^{(L)}\}$. Uncertainty on the pushforward modeled data samples $\rho_1(\mathbf{d})$, shown in Fig. 1,

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Algorithm 1 Sequential Monte Carlo (SMC) for pushforward

Input: a user-defined simple prior distribution $f_1(\boldsymbol{m})$, geophysical forward model $g(.)$, observed geophysical data $\mathbf{d}_{\text{obs}}$, number of Monte Carlo sequences N

Output: Monte Carlo samples of pushforward posterior data $\mathbf{d}^{(\ell)}$

At initial sequence time $i = 1$:

1. Estimate posterior $f_i(\mathbf{m} \mid \mathbf{d}_{obs})$ using Monte Carlo samples of $\mathbf{m}^{(\ell)} \sim f_i(\boldsymbol{m})$ and $\mathbf{d}^{(\ell)} = g\left(\mathbf{m}^{(\ell)}\right)$, obtaining the pushforward distribution $\rho_i(\mathbf{d})$.
2. Set $f_{i+1}(\mathbf{m}) = f_i(\mathbf{m} \mid \mathbf{d}_{obs})$ and $i = i + 1$.

At sequences $i \geq 2$:

3. Repeat Step 1 to Step 2, until $i = N - 1$:

At sequence $i = N$: set $f_i(\mathbf{m})$ as the proposal distribution $f_{prop}(\mathbf{m})$ and $\rho_i(\mathbf{d})$ as $\rho_{prop}(\mathbf{d})$:

4. Run final pushforward calculation (Eq (3)) to get $f_{post}(\mathbf{m})$. Return posterior samples: $\mathbf{m}^{(\ell)} \sim f_{post}(\boldsymbol{m})$ and $\mathbf{d}^{(\ell)} = g\left(\mathbf{m}^{(\ell)}\right)$

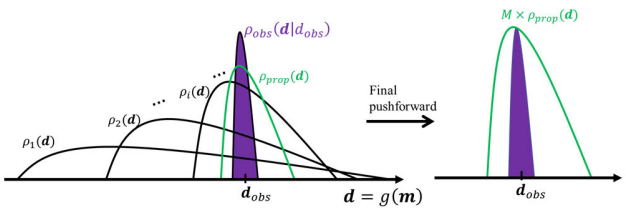

Fig. 1. Illustration of the proposed SMC, where M is the ratio between ρ_{obs} and ρ_{prop} (i.e., ρ_{push}) according to (3).

will thus be very wide Ω. Then, $f(\mathbf{m})$ and $\rho(\mathbf{d})$ will be updated through a sequence of intermediate distributions to reduce the uncertainty, until getting the final pushforward data distribution $\rho_{\text{push}}(.)$ that match the target distribution of observed data $\rho_{\text{obs}}(.)$.

There are two main reasons for choosing SMC.

1) Monte Carlo sampling leverages parallel computing. In addition, machine learning can be easily combined with Monte Carlo sampling to speed up convergence.
2) The prior distribution is not well known; hence the goal is to start with a simple and large prior and then sequentially narrow the prior in the sampling. In effect, the prior is really a proposal distribution that is updated sequentially to improve the sampling. The second reason is especially important to greenfield mineral exploration as geological information is limited for defining the prior distribution for inversion.

To estimate the $f_i(\mathbf{m} \mid \mathbf{d}_{\text{obs}})$ at each SMC sequence before pushforward, we use approximate Bayesian computation (ABC). ABC has the advantage of being likelihood-free, providing Bayesian inference solutions when the likelihood function is analytically intractable [46]. Such advantage makes it a natural choice for our geophysical inverse problem, particularly the real field application (see Section IV). By formulation, ABC approximates the posterior distribution using summary statistics near the observed data [47]

$$f_i(\mathbf{m}|\mathbf{d}_{\text{obs}}) \approx f(\mathbf{m}_i \mid S(\mathbf{d}_i, \mathbf{d}_{\text{obs}}) < \epsilon). \tag{9}$$

where $S(\mathbf{d}_i, \mathbf{d}_{\text{obs}})$ are the distance measure between the modeled data and actual measurements, such as L1 or L2 norms [48]. ϵ is acceptance tolerance. In implementation, we use ABC to estimate $f_i(\mathbf{m} \mid \mathbf{d}_{\text{obs}})$ distribution with the following steps summarized in Algorithm 2.

Algorithm 2 Approximate Bayesian Computation for $f_i(\mathbf{m} \mid \mathbf{d}_{obs})$

Input: Prior distribution $f_i(\mathbf{m})$ at SMC sequence i, observed geophysical data $\mathbf{d}_{\text{obs}}$, acceptance tolerance ϵ.

Output: Prior model distribution $f_i(\mathbf{m}|\mathbf{d}_{obs})$

1. Sample $\mathbf{m}^* \sim f_i(\mathbf{m})$, and forward model geophysical data $\mathbf{d}^* = g(\mathbf{m}^*)$
2. Compute $S(\mathbf{d}^*, \mathbf{d}_{obs})$.
3. Accept $\mathbf{m}^*$ as a posterior sample if $S(\mathbf{d}^*, \mathbf{d}_{obs}) < \epsilon$. Otherwise, reject $\mathbf{m}^*$.
4. Repeat above steps 1-3 until L posterior samples of $\mathbf{m}^*$ are accepted.
5. Use ABC accepted $\mathbf{m}^*$ samples to model $f_i(\mathbf{m}|\mathbf{d}_{obs})$

For modeling $f_i(\mathbf{m}|\mathbf{d}_{\text{obs}})$ using the ABC accepted samples, we can use kernel density estimation when the geological model $\mathbf{m}$ is in low dimension. When $\mathbf{m}$ is in high-dimensional, which are mostly the case for real fields, we can use machine learning density estimation approaches such as mixture of probabilistic principal component analyzers (MPPCAs) (Tipping and Bishop [49]) or Masked Autoregressive Flow (Papamakarios et al. [50]). In our case studies, we will use MPPCA and provide its implementation code. Another challenge for real fields is that the forward modeling of geophysical data $\mathbf{d}$ is time-consuming, especially when ABC requires a large sample size to accept sufficient posterior samples. We thus train a fast artificial neural network (ANN) surrogate model (see Appendix B) that directly predicts $\mathrm{S}(\mathbf{d}, \mathbf{d}_{\text{obs}})$, to replace the conventional geophysical modeling. Now, we can finalize our SMC sampler with ABC to solve the pushforward inversion.

The above algorithm runs in parallel for each Monte Carlo sample, making it computationally efficient. In the end, we obtain the pushforward samples $\mathbf{d}^{(\ell)}$ from geophysics forward modeling on the final posterior $\mathbf{m}^{(\ell)}$. We can then use the above-mentioned density estimation to obtain $\rho_{\text{push}}(.)$, and calculate the Bayes factor using (8). It is also important to note that the ANN surrogate won't introduce additional model errors as we use the full physics forward modeled $\mathbf{d}^{(\ell)}$ for calculating the Bayes factor.

III. 2-D Synthetic Case

We first verify the proposed approach on a simple 2-D synthetic case for gravity inversion. The observed gravity data is acquired over two high-density spherical bodies overlapping each other (Fig. 2). The bodies are buried in a homogeneous subsurface. As shown in the figure, the observed gravity anomalies ($\mathbf{d}_{\text{obs}}$) are acquired from between $x = -200$ and 200 m at surface. The measurement acquisition stations are spaced in 20 m. The goal is to recover the two dense bodies by inverting the observed gravity anomalies, similar to any typical

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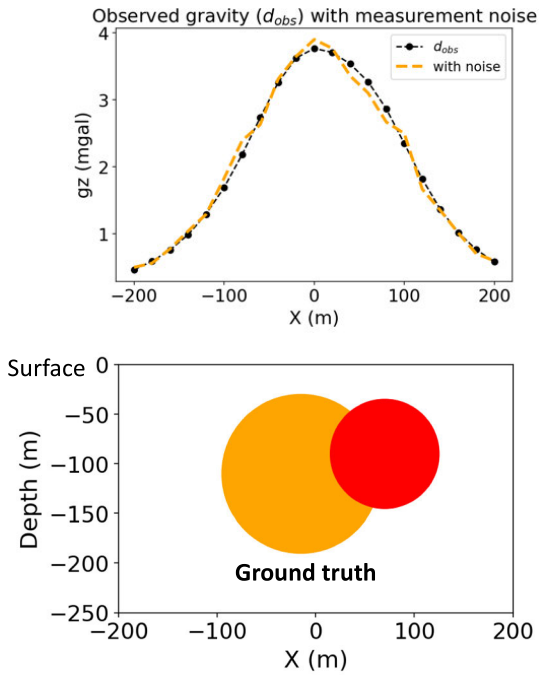

Fig. 2. Observed gravity data at the surface with the ground truth two-sphere dense body. Upper: the black-dotted line shows the noise-free observation data $\mathbf{d}_{\text{obs}}$. The dashed line is with 5% noise used for inversion. Lower: the density of each ground truth body is 2.61 g/cm^3 (left circle) and 2.71 g/cm^3 (right circle).

geophysical inversion study. Here, we replicate the realistic assumption that it is not known whether we have one, two, or more bodies.

A. Quantify Model Misrepresentation Errors in Gravity Inversion

1) Formulating the Misrepresentation Test: We invert the gravity data, starting by assuming one body is present and it is spherical (the simple assumption). The sphere model $\mathbf{m}$ is defined by three parameters: sphere radius (a), center depth (z), and density (rho): $\mathbf{m} = \{a, z, rho\}$. The sphere center is fixed at location of the signal peak where $x = 0$ m for simple illustration of the method (while in the following field study, we will vary the location parameters).

We use the above-defined hypothesis test and calculate the Bayes factor with the assumption of a single sphere model as the null hypothesis H_0. The Bayes factor for the hypothesis test at each gravity data station x is formulated as

$$B(x) = \frac{\sum \rho_{\text{push}}\left(\mathbf{d}_{\text{obs}}^{(\ell)}(x) | H_0\right)}{\sum \rho_{\text{obs}}\left(\mathbf{d}_{\text{obs}}^{(\ell)}(x)\right)} \tag{10}$$

where $\mathbf{d} = g(x, \mathbf{m})$ is the pushforward gravity data from the single sphere. We model the gravity signal g on a flat ground surface as a classic function of spatial location x (Blakely, 1995)

$$\mathbf{d} = g(x, \mathbf{m}) = \frac{4}{3} G\pi \frac{a^3 * rho * z}{\left(x^2 + z^2\right)^{3/2}} \tag{11}$$

where $G = 6.6743 \times 10^{-11}$ m$^3 \cdot$kg$^{-1} \cdot$s^{-2} is the gravitational constant. Both the ground truth and pushforward gravity data

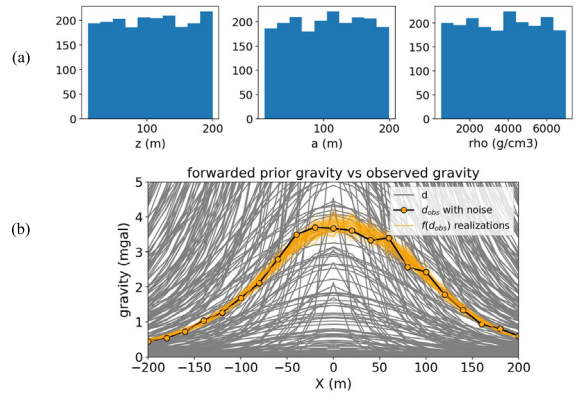

Fig. 3. (a) 2000 Monte Carlo samples of the prior parameter $\mathbf{m}$. (b) Corresponding samples of forward modeled gravity data $\mathbf{d}$ using prior parameters.

are calculated with the above forward model function, thus there are no physical model errors in the synthetic case.

We assign 5% of Gaussian measurement error to the observed gravity $\mathbf{d}_{\text{obs}}$, thus $f_{\text{obs}} = \mathcal{N}(\mathbf{d}_{\text{obs}}, (0.05\mathbf{d}_{\text{obs}})^2)$. Next, we run the SMC-ABC sampler to perform the gravity inversion and sample $\mathbf{d}^{(l)}$ from the posterior.

2) SMC Sampling From Gravity Inversion: The input prior is defined as a wide uniform distribution for model parameters, with $rho = \text{Uniform}(1500, 7000)$ kg/m^3, $a = \text{Uniform}(20, 200)$ m, and $z = \text{Uniform}(50, 200)$ m. We run the Monte Carlo sampling from the prior model distribution and forward model the prior gravity data. A total of 2000 realizations are generated (Fig. 3).

The implemented SMC-ABC sampler of Algorithm 3 is then called to perform the gravity inversion, with the prior samples as input. A python notebook[1] is provided in our GitHub to illustrate its implementations on the synthetic case. We ran two SMC-ABC sequences before the final pushforward. Fig. 4(a) shows the model parameter distribution of SMC-ABC sequence, with the pdf of $f(\mathbf{m})$ modeled by MPPCA. The ABC keeps samples $\mathbf{m}$ from the modeled $f(\mathbf{m})$ and only accepts when the ANN predicted misfit smaller than the threshold [Fig. 4(b)]. The ABC tolerance ϵ is set as 2.5th percentile (P2.5) of the mismatch from prior samples, and we applied the same rule for the following field study. In the end, 2000 proposed samples ($L = 2000$) are accepted for the pushforward.

Fig. 5 shows the final pushforward posterior samples and compares them to the prior. We notice a significant uncertainty reduction in the posterior compared to the prior in Fig. 3(a). The obtained posterior ensemble also recovered a nonlinear correlation between density rho and radius a. The forwarded posterior data clearly does not match the gravity data between 40 and 120 m [Fig. 6(a)]. Fig. 6(b) further plots the ensemble of posterior sphere bodies, showing the single posterior sphere cannot reproduce the second smaller sphere (colored in red). Next, we show how this obvious problem is detected with a hypothesis test and how the Bayes' factor provides insight into the problem of a too simple model.

3) Model Misrepresentation Test and Error Reduction: We calculate the Bayes factor using the SMC-ABC sampled

[1]https://github.com/sdyinzhen/Model-Error-Quantification/tree/main

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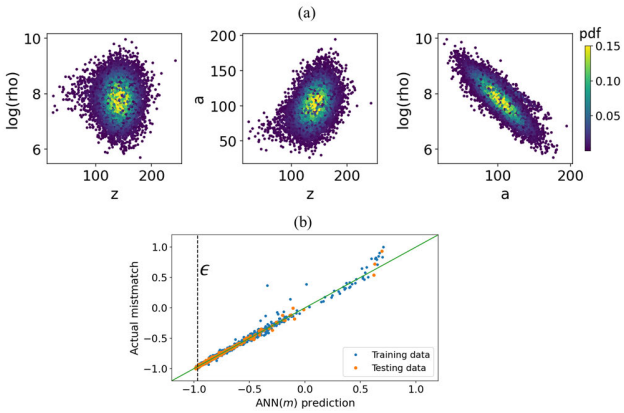

Fig. 4. (a) Probability density of the sphere parameter $\mathbf{m}$ modeled by MPPCA. (b) Prediction performances of the training and test data using the ANN surrogate model.

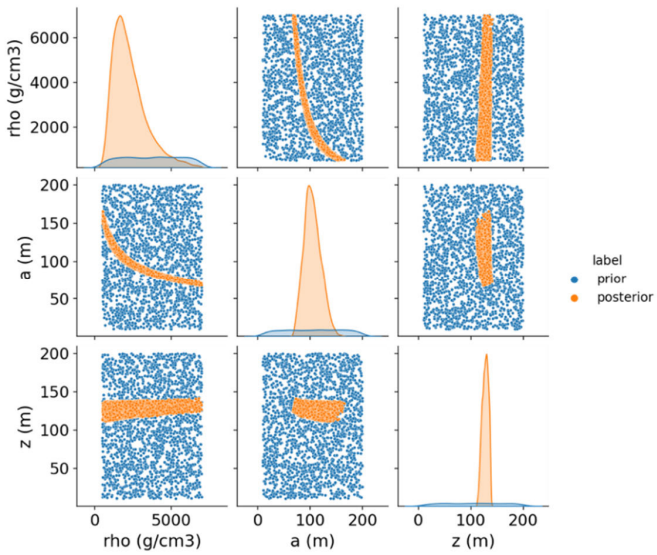

Fig. 5. Posterior model parameters versus initial wide prior for geological model with a single sphere dense body.

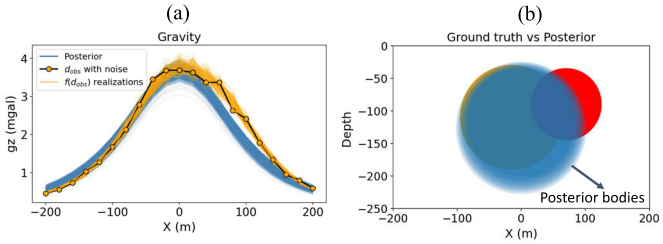

Fig. 6. (a) Forward modeled posterior gravity data (blue lines) compared to the noisy (black line) used for inversion. The brown lines ("$f(\mathbf{d}_{\text{obs}})$ realizations") visualize $f(\mathbf{d}_{\text{obs}})$ distribution. (b) Visualization of the ensemble posterior dense bodies comparing to the ground truth.

posterior data with (10). The resulting Bayes factor is plotted in Fig. 7(a). We expect the Bayes factor to be close to 1 in the case of no or negligible model error, or, the log of the factor ($\log(B)$) close to zero. In the figure, we notice very small Bayes factors (large negatives in the log) values at gravity stations between 20 and 120 m ($\log(B) \ll 0$), indicating substantial misrepresentation errors. The sphere model hypothesis is falsified. However, from this one may

Algorithm 3 SMC-ABC Sampler With Neural Network Surrogate

Input: User-defined simple prior $f_1(\mathbf{m})$ of model parameters under hypothesis H_1, forward model$g(.)$, observed geophysical data $\mathbf{d}_{\text{obs}}$, acceptance tolerance ϵ, number of SMC sequences N.

Output: Posterior samples $\{\mathbf{d}^{(1)}, \ldots, \mathbf{d}^{(\ell)}, \ldots, \mathbf{d}^{(L)}\}$ that match $\mathbf{d}_{\text{obs}}$.

1. Initialize with Monte Carlo sampling of prior: $\mathbf{m}_1^{(\ell)} \sim f_1(\mathbf{m})$, where $\ell = 1, 2, \ldots, \text{L}$. Forward model geophysical data: $\mathbf{d}_1^{(\ell)} = \text{g}\left(\mathbf{m}_1^{(\ell)}\right)$. Compute the misfit with L2 norm: $\text{S}_1^{(\ell)} = \left\|\mathbf{d}_1^{(\ell)}, \mathbf{d}_{\text{obs}}\right\|_2$.
2. Train ANN surrogate model using the Monte Carlo samples of $\left(\mathbf{m}_1^{(\ell)}, \text{S}_1^{(\ell)}\right)$, obtaining $\text{S}_1 = \text{ANN}(\mathbf{m}_1)$.

For $i = 1, \ldots, \text{N} - 1$:

3. Sample $\mathbf{m}_i^* \sim f_i(\mathbf{m})$, accept when $\text{ANN}(\mathbf{m}_i^*) < \epsilon$. Repeat until accepting L samples, defining the accepted samples as $\mathbf{m}_{i+1} = \left\{\mathbf{m}_{i+1}^{(1)}, \mathbf{m}_{i+1}^{(2)}, \ldots, \mathbf{m}_{i+1}^{(\text{L})}\right\}$.
4. Forward model geophysical data $\mathbf{d}_{i+1}. = g(\mathbf{m}_{t+1}.)$ and compute $\text{S}_{i+1} = \left\|\mathbf{d}_{i+1}^{(\ell)}, \mathbf{d}_{\text{obs}}\right\|_2$.
5. Model the new sequence of distribution $f_{i+1}(\mathbf{m})$ from the accepted samples of $\mathbf{m}_{i+1}$
6. Update the ANN surrogate with $\left(\mathbf{m}_{i+1}^{(\ell)}, \text{S}_{i+1}^{(\ell)}\right)$, obtaining $\text{S}_{i+1} = \text{ANN}(\mathbf{m}_{i+1})$
7. Set $i = i + 1$.

End

8. Set $\left\{\mathbf{m}_N^{(1)}, \mathbf{m}_N^{(2)}, \ldots, \mathbf{m}_N^{(\text{L})}\right\}$ as $\mathbf{m}_{prop}$, and forward model data $\mathbf{d}_{prop} = g\left(\mathbf{m}_{prop}.\right)$
9. Run pushforward using (3) with $\mathbf{m}_{prop}$ and $\mathbf{d}_{prop}$, obtaining $f_{post}(\mathbf{m})$.
10. Return posterior samples: $\mathbf{m}^{(\ell)} \sim f_{post}(\mathbf{m})$ and $\mathbf{d}^{(\ell)} = g\left(\mathbf{m}^{(\ell)}\right)$

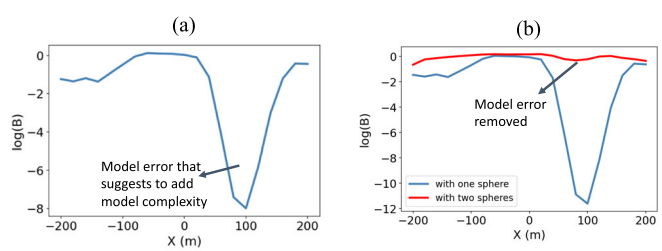

Fig. 7. (a) Bayes factor for the posterior with single sphere. (b) Bayes factor for the posterior using two spherical bodies.

deduce that model complexity needs to increase where the log Bayes factor is smallest, for example by adding one more sphere to the model. Therefore, we add another sphere body to the center of falsified locations ($x = 80$ m).

We then rerun the misrepresentation test by repeating the SMC-ABC inversion procedures but for the two spherical bodies. For the input prior, we keep the first sphere parameters $\{a_1, z_1, rho_1\}$ unchanged [Fig. 3(a)], and simply define the prior distribution of the second sphere parameters $\{a_2, z_2, rho_2\}$ same as the first. Details of the SMC-ABC

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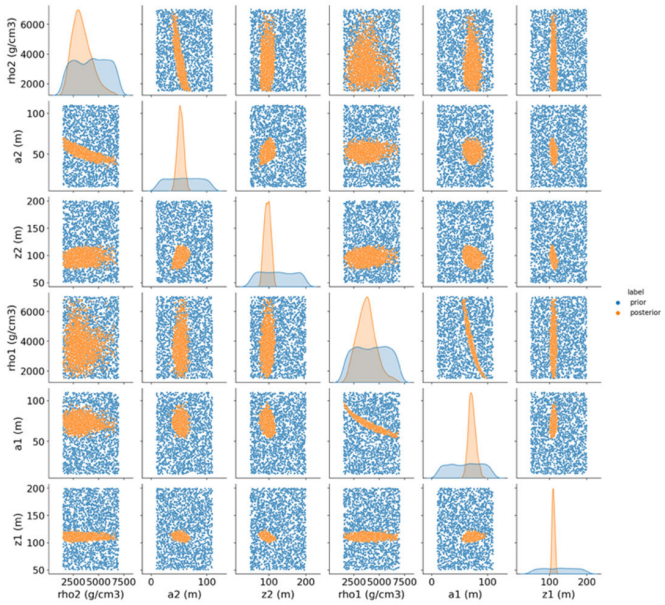

Fig. 8. Posterior model parameters versus initial wide prior when modeling using two sphere dense bodies.

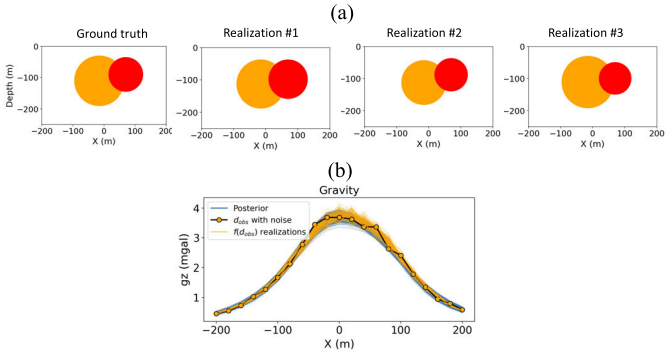

Fig. 9. (a) Posterior model realizations after adding another sphere. (b) Forward modeled gravity with two spheres compared $\mathbf{d}_{\text{obs}}$.

inversion with the two spheres are also demonstrated in the same Python notebook[2].

Fig. 8 plots the newly inverted posterior results. The posterior parameters show the two spherical body parameters have no correlations between each other. Fig. 9(a) plots the spheres using posterior parameters and compares them to the ground truth, where the second smaller sphere is correctly reproduced. The forward modeled posterior data now match the observed gravity at all stations [Fig. 9(b)], with much higher matching quality than using the single sphere. As a result, the log Bayes factor is close to zero for all stations in Fig. 7(b). The model errors have disappeared when compared to Fig. 7(a), meaning the two-sphere model hypothesis cannot be rejected.

In summary, the misrepresentation test using the Bayes factor detects and measures the model errors of the synthetic inversion study. This approach offers a quantifiable criterion to stochastically assess and adaptively revise the geological model hypothesis. Through mitigating quantified model errors, the model representation converged toward the synthetic ground truth geology. The SMC sampler was shown to be an effective tool for generating the inverted posterior geological models, hence enabling practical implementation of the hypothesis test. It allowed the gravity inversion to be performed by inputting an initial wide prior distribution of model parameters.

The synthetic implementation, executed on a single PC (2.4-GHz Intel i9 CPU, 32-GB RAM), took only less than 5 min. We have made the codes with SMC sampler, and the case study open-source through a Python notebook[3] to assist the public in their practical implementations.

IV. Application to the Real-Field Critical Mineral Exploration

A. Geological Background and Field TDEM Surveys

We apply the synthetic tested approach to an actual field exploration searching for nickel-sulfide deposits in the Mid-continent Rift system. The prospective critical mineral deposits are hosted in a large, mafic–ultramafic intrusion [34], where the deposits are likely to be formed at the bottom of intrusions. No drillholes intersecting the presented targets are available at this stage. A set of ground loop time domain electromagnetic (TDEM) surveys was acquired over the field. Fig. 10(a) shows the maps of acquired TDEM field data $\mathbf{d}_{\text{obs}}$. As a spatial dataset with time series, the acquired TDEM map has a total of 12 time channels, from 240 to 880 ms. The TDEM signal decays over time, as shown in the figure. Additionally, the TDEM is a vector field dataset that measures the magnetic field in three directions: horizontal directions X and Y, and vertical Z. The TDEM map in Z-direction, $\mathbf{d}_{\text{obs}}(Z)$, is consistent with three buried conductor anomalies (most clearly visible as three peaks in the early time-channel data: 240.86, 299.02 ms) in Fig. 10(a). These conductor anomalies are candidates for highly conductive massive sulfide orebodies. The goal of the field study is to use the acquired TDEM dataset to invert the conductors' depth, sizes, and potential volumes. Quantifying the uncertainty of these parameters is substantially important because they directly impact exploration strategies.

B. TDEM Inversion With Simple Geological Models

No direct information on the shape of the hypothesized conductors is available. Some typical shapes of massive sulfide deposits are illustrated in the literature [34], [51]. A simple initial model to begin the analysis of the TDEM data is to use three 3-D spherical conductors [Fig. 10(b)]. Each spherical conductor is defined by five parameters: conductor radius (size) r, conductor center location X, Y, depth Z, and conductivity σ. Three conductors require a total of 15 parameters. Representing the complex geology with simple 3-D spheres is an oversimplification, but the key question here is to quantify the model misrepresentation, such that additional geometrical features can be added.

We use the acquired TDEM data to test the three-sphere conductor hypothesis. Following the proposed methodology, the Bayes factor is formulated with the TDEM as test data

$$B(\boldsymbol{x}, t) = \frac{\sum \rho_{\text{push}}\left(\mathbf{d}_{\text{obs}}(\boldsymbol{x}, t)^{(\ell)} \mid H_0\right)}{\sum \rho_{\text{obs}}\left(\mathbf{d}_{\text{obs}}(\boldsymbol{x}, t)^{(\ell)}\right)}. \tag{12}$$

[2]https://github.com/sdyinzhen/Model-Error-Quantification/tree/main [3]https://github.com/sdyinzhen/Model-Error-Quantification/tree/main

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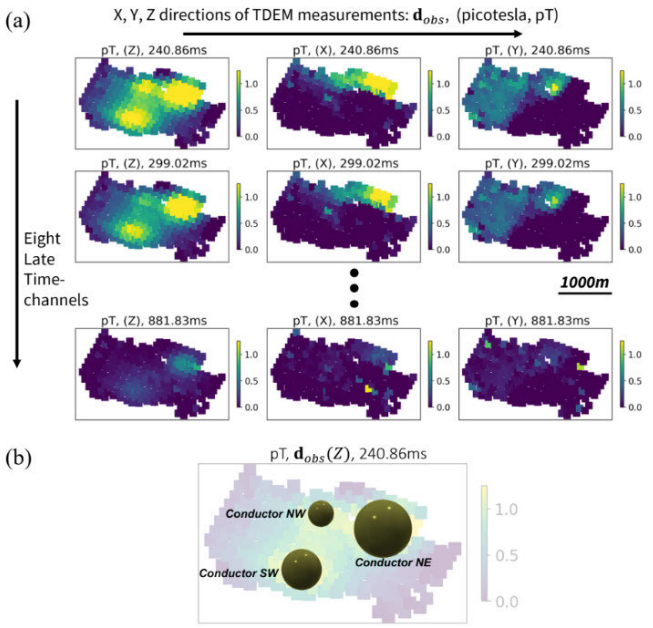

Fig. 10. (a) Maps of the real field airborne TDEM surveys. (b) Anomalies in the Z-direction correspond to three hypothesized conductors beneath the surface.

Here, we define the three spherical conductors as null hypothesis H_0. $\mathbf{m}$ are the 15 model (3-D spherical) parameters under H_0. $\rho_{\text{push}}(.)$ is the distribution of pushforward posterior TDEM data. For the TDEM pushforward model $g(.)$, we use SIMPEG, an open-source geophysical modeling package [52]. SIMPEG solves the quasi-static formulation of Maxwell's equations on a meshed conductivity model to obtain the time-dependent resulting magnetic fields in response to a pulsed current from a large ground loop.

The SMC-ABC sampler is run to sample the posterior $\mathbf{d}(\boldsymbol{x}, t)^{(\ell)}$ for $\rho_{\text{push}}(.)$, using the provided open-source codes but with use of SIMPEG to calculate $g(.)$. Table I summarizes the initial intentionally wide prior parameter distribution of the three spherical conductors. The electrical conductivity is defined as a log-normal distribution. The mean and standard deviation are decided using well-log data from nearby boreholes in the same intrusion.

We generate 2000 Monte Carlo realizations from the output SMC-ABC pushforward posterior. The posterior model parameters are shown in Fig. 11(a). We observed uncertainty reductions for all the 15 parameters when compared to the prior. Interestingly, the conductor NE with the strongest TDEM signals is found in this inversion to be likely smaller in size and shallower in depth, while the other two conductors appear to be larger and deeper.

Fig. 12 shows three realizations of the posterior TDEM ($t = 240.86$ ms) data $\mathbf{d}(\boldsymbol{x}, t)^{(\ell)}$. We observe they largely match the observed field measurements in X-, Y-, and Z-directions of the magnetic field. To examine the entire 2000 posterior realizations, we take a 1-D cross section (A-A′) and make the ensemble plot shown in Fig. 13(a). The 2000 posterior matched the observed TDEM, at both first (240.86 ms) and second (299.02 ms) time channels, with a few mismatched regions (pointed by the red arrows) that need to be improved.

TABLE I

PRIOR SPHERE MODEL PARAMETER $\mathbf{m}$ AND ITS PRIOR UNCERTAINTY DISTRIBUTION $f(\mathbf{m})$. THE PRIOR DISTRIBUTIONS OF THE PARAMETERS ARE MOSTLY ASSUMED TO BE UNIFORM (FORMULATED AS U[min, max]) DUE TO LIMITED INFORMATION

Model	Global parameters: m	Prior uncertainty: f(m)	Source for prior uncertainty statement
Conductor NE	Conductor radius, r_{NE}	U[50, 450] meter	Initial wide prior relative to the EM anomaly; well log data from nearby boreholes in the same intrusion.
	Center X, X_{NE}	U[-600, 600] meter	
	Center Y, Y_{NE}	U[-600, 600] meter	
	Depth, Z_{NE}	U[-90, -690] meter	
	Conductivity, σ_{NE}	Lognormal(1.6, 0.8^2)	
Conductor SW	Conductor radius, r_{SW}	U[50, 350] meter	
	Center X, X_{SW}	U[-650, 650] meter	
	Center Y, Y_{SW}	U[-700, 700] meter	
	Depth, Z_{SW}	U[-50, -600] meter	
	Conductivity, σ_{SW}	Lognormal(2.2, 0.8^2)	
Conductor NW	Conductor radius, r_{NW}	U[50,350] meter	
	Center X, X_{NW}	U[-600, 600] meter	
	Center Y, Y_{NW}	U[-650, 650] meter	
	Depth, z_{NW}	U[-60, -660] meter	
	Conductivity, σ_{NW}	Lognormal(1.6, 0.8^2)	

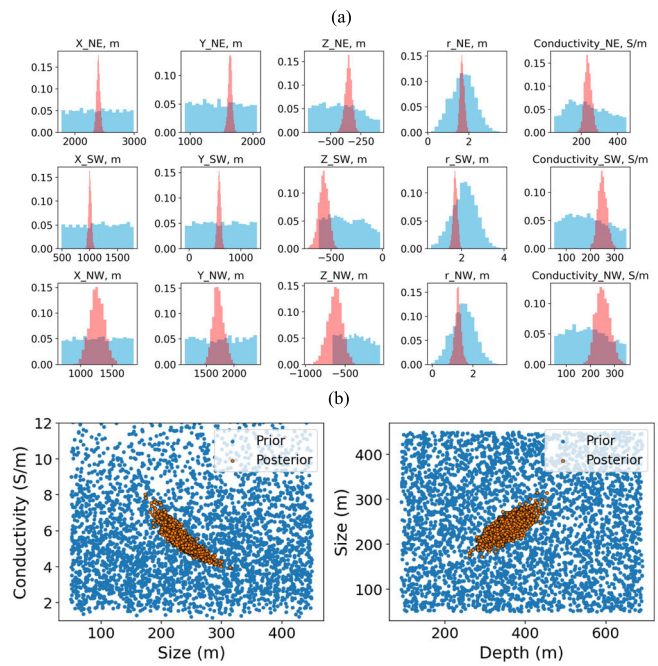

Fig. 11. (a) Posterior sphere parameters. (b) Correlations between model parameters (size, conductivity, and depth) for conductor NE that has the strongest TDEM signals.

Note that the first two channels are the most difficult to match, as they show the anomalies of all the three conductors.

C. Testing Misrepresentation Errors

We can now compute the Bayes factor to answer our central question regarding the model misrepresentation from sphere conductors. The Bayes factors are computed at every location of the testing TDEM data, thus creating the misrepresentation maps shown in Fig. 14(a). We observe most areas have high Bayes factor values (lighter color) in log scale, especially at the SW conductor. It suggests the sphere is not too bad to start with for modeling the geological conductor. However, the log Bayes' factors are especially low (darker color) for

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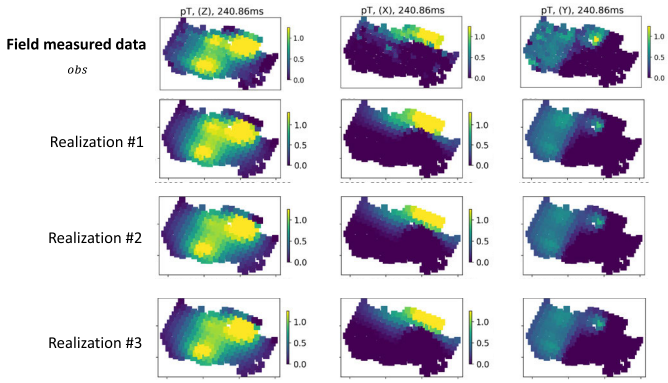

Fig. 12. Forward modeled TDEM realizations from the 3-D spherical posterior versus the field observed TDEM data.

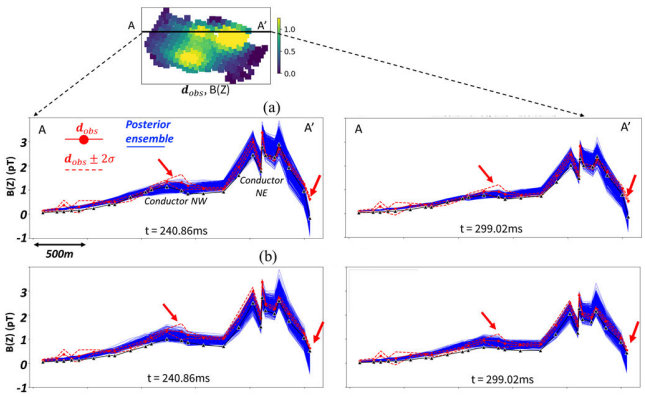

Fig. 13. Ensemble plot of the posterior TDEM data in 1-D at cross section A-A′. (a) With sphere geological models. (b) With ellipsoid geological models.

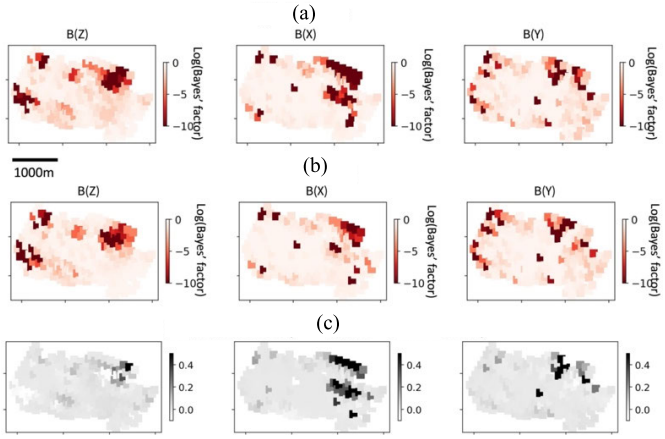

Fig. 14. Maps of Bayes factor in log scale (a) sphere conductor models and (b) ellipsoid conductor models. (c) Bayes factor difference between sphere and ellipsoid models.

the NE conductor. Since the TDEM signals are the strongest at conductor NE, the low Bayes factor means the hypothesis of the 3-D sphere conductor is very likely to be wrong. Low Bayes factor values are also observed at conductor NW. All these findings in the misrepresentation error are difficult to detect when visually inspecting the data matching qualities in Figs. 12 and 13(a).

TABLE II

PRIOR ELLIPSOID MODEL PARAMETER **m** AND ITS PRIOR UNCERTAINTY DISTRIBUTION $f(\mathbf{m})$

Model	Global parameters: m	Prior uncertainty: f(m)	Source for prior uncertainty statement
Conductor NE	Ellipsoid major axis, a_{NE}	U[50, 450] meter	Initial wide prior relative to the EM anomaly; well log data from nearby boreholes in the same intrusion.
	Major and median axes ratio, $Rt_{NE}^{maj/med}$	U[0.2, 1]	
	Median and minor axes ratio, $Rt_{NE}^{med/min}$	U[0.2, 1]	
	Heading, $\emptyset_{NE}$	[-90, 90]Deg	
	Strike, θ_{NE}	[-90, 90]Deg	
	Roll, ψ_{NE}	[-90, 90]Deg	
	Center X, X_{NE}	U[-600, 600] meter	
	Center Y, Y_{NE}	U[-600, 600] meter	
	Depth, Z_{NE}	U[-90, -690] meter	
	Conductivity, σ_{NE}	Lognormal(1.6, 0.8^2)	
Conductor SW	Ellipsoid major axis, a_{SW}	U[50, 350] meter	
	Major and median axes ratio, $Rt_{NE}^{maj/med}$	U[0.2, 1]	
	Median and minor axes ratio, $Rt_{NE}^{med/min}$	U[0.2, 1]	
	Heading, $\emptyset_{SW}$	[-90, 90]Deg	
	Strike, θ_{SW}	[-90, 90]Deg	
	Roll, ψ_{SW}	[-90, 90]Deg	
	Center X, X_{SW}	U[-650, 650] meter	
	Center Y, Y_{SW}	U[-700, 700] meter	
	Depth, Z_{SW}	U[-50, -600] meter	
	Conductivity, σ_{SW}	Lognormal(2.2, 0.8^2)	
Conductor NW	Ellipsoid major axis, a_{NW}	U[50, 350] meter	
	Major and median axes ratio, $Rt_{NE}^{maj/med}$	U[0.2, 1]	
	Median and minor axes ratio, $Rt_{NE}^{med/min}$	U[0.2, 1]	
	Heading, $\emptyset_{NW}$	[-90, 90]Deg	
	Strike, θ_{NW}	[-90, 90]Deg	
	Roll, ψ_{NW}	[-90, 90]Deg	
	Center X, X_{NW}	U[-600, 600] meter	
	Center Y, Y_{NW}	U[-650, 650] meter	
	Depth, z_{NW}	U[-60, -660] meter	
	Conductivity, σ_{NW}	Lognormal(1.6, 0.8^2)	

Next, we revise the model representation by using ellipsoids. A 3-D ellipsoid has more freedom to represent model complexity than a sphere. Its shape can vary independently in three directions (major, median, and minor axes), and it can rotate freely in three independent angles (heading, strike, and roll). Spheres are included as a subset of ellipsoids. Appendix C shows details of the ellipsoid model parameterizations. Each ellipsoid conductor has a total of ten parameters. Three conductors will have 30 independent parameters, in much higher dimensions than the previous sphere models with 15 parameters.

We will need to retest the model errors from the ellipsoid. Table II shows the input prior distribution of the 30 model parameters under the ellipsoid model hypothesis, mostly as uniform distributions except for log-normal for conductivity. Specifically, we set the semi-major axis of each conductor the same as the radius of the previous spheres model. The median and minor axes are defined as a fraction of the major and median axes, respectively. Each fraction is uniformly distributed between 0.2 and 1. Each rotation angle will be between -90 and $+90°$, allowing the ellipsoid to fully rotate in the entire space. The prior distribution of X, Y, depth, and conductivity are kept the same as the previous sphere.

D. Results: Inversion With Improved Model Complexity

Fig. 15 shows the posterior results from the SMC-ABC inversion. Similar to the sphere model, substantial uncertainty reductions from the prior are obtained for all the conductors. High levels of data matching quality with the observed TDEM are shown in Figs. 13(b) and 16. Furthermore, the ellipsoid

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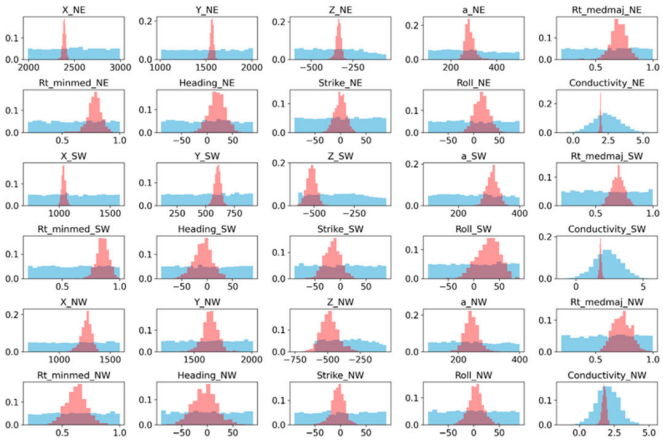

Fig. 15. Prior versus posterior ellipsoid parameters.

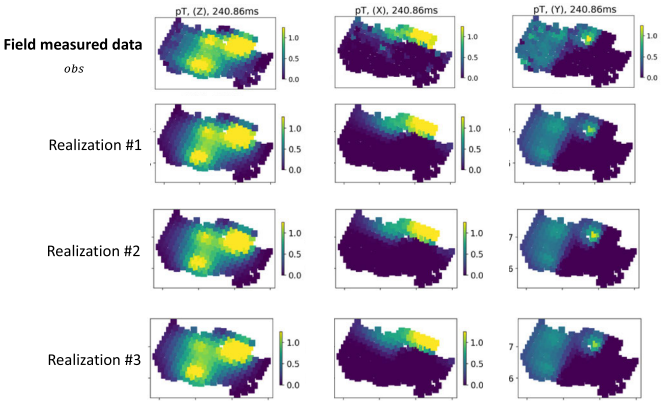

Fig. 16. Forward modeled TDEM from the ellipsoid posterior versus the field observed TDEM data.

posterior shows better-improved matching results at signal anomaly locations where some sphere models failed to match (pointed out by the red arrows in Fig. 13). Did the 3-D ellipsoid reduce the model errors and prove to be a better representation of the conductors We test the new model hypothesis by calculating the Bayes factor. Fig. 14(b) shows the results. Compared to the sphere Bayes factor in Fig. 14(a), there is an overall drop in the values in Fig. 14(b). On average, the Bayes factor values on the entire map dropped by 24%, 19%, and 4% in Z-, X-, and Y-directions, respectively. The dropping levels are the biggest for conductor NE and NW areas. They indicate the reduction of model misrepresentation errors.

We can use the posterior model to evaluate the conductor volume, a key economic parameter concerning field development. Fig. 17 shows the posterior conductor volumes from the sphere and ellipsoid. They are remarkably different, although their forwarded data matched the observed TDEM similarly. Ellipsoid was shown to have much less volume uncertainty than the sphere. Especially for the NW conductor, the sphere model suggests it can have extremely large volumes, though its TDEM signal decays the fastest. Its volume uncertainty is also very big, varying between $0.25 \times 10^8 - 1.54 \times 10^8$ m^3. The ellipsoid helped to avoid this problem with much smaller volumes and much less uncertainty. Less uncertainty in volume

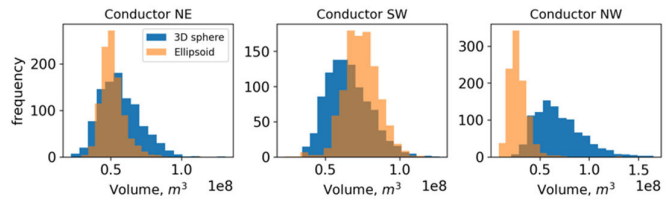

Fig. 17. Posterior conductor volumes: using 3-D sphere versus using ellipsoid.

will help to reduce the risk and give more confidence for the field development decisions.

The contrast in posterior volume uncertainty underscores the importance of quantifying misrepresentation errors in geophysical inversion. We understand ellipsoid models are still simple to represent the actual geology, and moving beyond simple ellipsoids will keep improving model fits (and associated Bayes factor). Indeed, the Bayes factor further reveals remaining high errors at specific locations as shown in Fig. 14(b). We can keep improving model complexity by iteratively repeating the above procedures. For instance, we can add a second ellipsoid to the conductor anomalies. As no other geological information such as borehole data is available, the misrepresentation error testing provides a tool for guiding the inference of a meaningful geological model for decision-making.

V. Discussion and Conclusion

We introduced a comprehensive stochastic treatment for model misrepresentation errors. These errors are demonstrated to be one major source of uncertainty in geophysical inversion, particularly at the early stage of greenfield explorations as evidenced by the synthetic and real field studies. To test whether the misrepresentation errors exist, we provided a generalized hypothesis testing using the geophysical surveys as testing data. The testing approach attempts to falsify the current model representation hypothesis by computing a Bayes factor. Sambridge et al. [53] also used a similar evidence ratio to the Bayes factor to evaluate the number of unknown model parameters in inversion with a linear Gaussian process. But here, we focus on quantifying the model misrepresentation errors. The Bayes factor measures the misrepresentation errors, both spatially and temporally, as the evidence ratio between the modeled data and actual observation. In the end, the combination of geophysical inversion with the hypothesis testing and Bayes factor provided a recursive inversion-testing-learning-improving process. This process allowed the iterative improvement of geological model representations. We applied the proposed misrepresentation treatment in two cases: a designed generic inversion project with gravity data, and a real greenfield project with TDEM data for nickel-sulfide explorations. Both the synthetic and real field studies showed the Bayes factor can accurately locate the misrepresentations and quantify their magnitudes. The real field application further showed that, through quantifying and then mitigating the measured misrepresentation errors, we reduced the uncertainty in key parameters concerning economic decision-making, including the conductor's volume

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and depth. As the misrepresentation errors haven't been intensively investigated in previously published studies, our study contributed a unique tool for the error testing and quantifying the inversion model uncertainty.

Another benefit of our approach is that it enables geophysical inversion to start with simple geological models, such as spheres, that are known to be overly simplified. Then the calculated misrepresentation errors allow to learn and correct the geometry. The pushforward formulation of the geophysical inverse problem keeps the model errors separated from data errors. The proposed SMC sampler for pushforward further allowed inputting a simple prior distribution, such as uniform distribution, of the simple model parameters. The SMC was shown to efficiently sample the posterior realizations that matched the observed geophysical data. The sampled posterior also retained correlations between the model parameters, either linear or nonlinear. All the above enabled the practical calculation of the Bayes factor. The simplified requirement on the input also facilitated easy implementation in complex real fields, without being restricted by the absence of prior geological information. We further provided detailed implementation algorithms alongside the open-source demonstration codes to facilitate practical applications.

Appendix

A. Derivation of the Data-Space Bayes Factor

The Bayes factor is classically defined as

$$B(H_0, H_1) := \frac{P(\mathbf{d}_{\text{obs}}|H_0)}{P(\mathbf{d}_{\text{obs}}|H_1)}. \tag{A.1}$$

Usually, (A.1) is marginalized over m to obtain (2). However, if we instead introduce a new random variable $\mathbf{d}$ that represents data from either measurements or simulations. Note that $\mathbf{d}$ and $\mathbf{d}_{\text{obs}}$ are different variables [54]. Now we can marginalize over $\mathbf{d}$ such that

$$\begin{aligned} P(\mathbf{d}_{\text{obs}} \mid H) &= \int_{\Omega_{\mathbf{d}}} \rho(\mathbf{d}_{\text{obs}}, \mathbf{d} \mid H) \mathrm{d}\mathbf{d} \\ &= \int_{\Omega_{\mathbf{d}}} \rho(\mathbf{d}_{\text{obs}} \mid \mathbf{d}, H) \rho(\mathbf{d}|H) \mathrm{d}\mathbf{d}. \end{aligned} \tag{A.2}$$

We now recognize that $\rho(\mathbf{d}_{\text{obs}} \mid \mathbf{d}, H) \equiv \rho_{\text{obs}}(\mathbf{d})$ and $\rho(\mathbf{d} \mid H) \equiv \rho_{\text{push}}(\mathbf{d} \mid H)$. Combining everything together, we obtain (4)

$$B(H_0, H_1) = \frac{\int_{\Omega_{\mathbf{d}}} \rho_{\text{push}}(\mathbf{d} \mid H_0) \rho_{\text{obs}}(\mathbf{d}) \mathrm{d}\mathbf{d}}{\int_{\Omega_{\mathbf{d}}} \rho_{\text{push}}(\mathbf{d} \mid H_1) \rho_{\text{obs}}(\mathbf{d}) \mathrm{d}\mathbf{d}}. \tag{A.3}$$

B. ANN Surrogate Model

We use a deep neural network to predict the geophysical data misfit directly using the conductor model parameters (Fig. 18). This is to reduce the geophysical forward modeling computation time when running a large ensemble of model realizations for the ABC.

In implementation, we used the classical Keras [55] python library to create the ANN surrogate, as Keras enables easily adding layers with high-level automation, and it is easy to implement. For the loss function, the mean squared error is used. The Adam optimizer [56] is then used to find the best weight parameters for the neural networks. Fig. 4(b) evaluates the model using both the training and testing set. Codes are provided in this article.

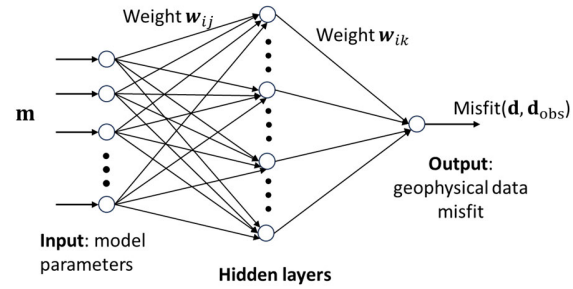

Fig. 18. Architecture of the ANN surrogate for predicting geophysical data misfit.

C. Parameterization of an Ellipsoid

An ellipsoid conductor can be represented by a total of ten parameters.

1) *a, b, c:* Three length parameters of the semi-axes in major, median, and minor directions. In the field implementation of Table II, we used the axes ratios $Rt^{\text{med/maj}}$ (between b and a) and $Rt^{\text{min/med}}$ (between c and b). This is to make sure that the major axis is larger than median, and median is larger than minor axis.
2) *Center X, Y, Z:* Ellipsoid center location in X-, Y-, and Z- (depth) directions.
3) *Heading $\emptyset$, Strike θ, Roll ψ:* Three independent rotation angles along Z-, Y-, and X-axes, respectively.
4) *Conductivity σ:* Conductivity of the ellipsoid conductor.

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